VIVAAN KUMAR GHOSAL

Aspiring Computer Science Engineer

[Portfolio] Kolkata, India

SUMMARY

IB Diploma candidate with a strong foundation in Physics, Mathematics and Computer Science, and hands-on experience across robotics, UAV design, machine learning and biotechnology. National-level footballer who represented Team India at the FC Barcelona Academy World Cup, and a school leader in sport, debate and science.

EDUCATION

International Baccalaureate (IB) Diploma Programme *May '27*
Modern High School International (MHSI), Kolkata, India
Higher Level: Physics, Chemistry, Mathematics: Analysis & Approaches | Standard Level: Economics, German B, English A
Predicted Score: [XX/45]

Grade 9, 10 (Cambridge IGCSE) *[Mon 'YY]*
Modern High School International, Kolkata, India
3 A*s and A grades in all other subjects except French; Cambridge ICE Distinction. Final Grade: [XX%]

SKILLS

Technical & Engineering: Computer Vision, Machine Learning, Arduino, PID Control, UAV Design, [Programming languages – confirm]

Coursework & Certifications: Plaksha University Young Technology Scholars (2025); IB HL Physics, Chemistry, Mathematics; Cambridge IGCSE (Computer Science, French); [Other online courses / certifications]

Languages & Music: English, German (IB SL), French (IGCSE), [Hindi / Bengali – confirm]; Drums (Trinity Grade 4, Distinction), Guitar

ACADEMIC AND RESEARCH PROJECTS

Low-Cost Fixed-Wing UAV for Hyperspectral Crop-Health Imaging *[Mon 'YY] – Present*
Robotics & precision agriculture project; provisional patent pending

- Designed a fixed-wing UAV for hyperspectral crop-health imaging at roughly 1/30th–1/40th the cost of industry-standard systems.
- Presented data in a format accessible to non-literate farmers, aiming to reduce pesticide overuse.
- Reached the Top 6 of 350+ teams (including international and funded teams) at the IIT-Kharagpur Young Innovators Programme.

GPS-Denied Autonomous UAV for Level Flight *[Mon 'YY]*
Computer vision and machine learning project

- Built a correcting plane that maintains level flight using computer vision and machine learning (Otsu's binarization), without GPS.
- Presented the work at the Get Set Learn national educators' conference.

Non-Destructive Testing of 3D-Printed Polymers *[Mon 'YY] – Present*
Long-term independent research paper

- Investigating the internal microstructure of 3D-printed polymers using light-based, non-destructive testing.
- Developed a machine learning program that analyzes incident and reflected rays and maps their paths, replacing manual data collection.

Force Plate and Stabilogram System *[Mon 'YY] – Present*
Robotics portfolio project

- Building a low-cost force plate and stabilogram system that measures fluctuations in centre of pressure (COP) using an Arduino-based setup, to help detect neural issues related to muscular control early.

Community Forum: Multilingual Civic-Communication Website *[Mon 'YY]*
Goethe-Institut Pune Youth Leaders Congress

- Built a multilingual website bridging the communities of Pune and the local municipality to improve communication between residents and government.
- Received an honorary award from a German Consulate representative at the Goethe-Institut Pune Youth Leaders Congress.

Sound-to-Electricity Energy Conversion *2023*
Cambridge Science Competition
- Developed a project converting sound energy into electricity; awarded a Silver Certificate in the Cambridge Science Competition 2023.

DNA Extraction Research *[Mon 'YY]*
Plaksha University, under Dr. Arshdeep Sidhu
- Extracted DNA from white blood cells and analyzed the results to deepen knowledge of biotechnology.

Hands-On Engineering Projects *[Mon 'YY] – Present*
- Built a six-gear and reverse LEGO car from scratch, without instructions, to study gear ratios.
- Contributing to the repair of an old Rover Ambassador, including gearbox extraction, piston repair and cleaning off dysfunctional wiring; started building a six-legged hexapod walking robot.
- Programmed and built a combat robot for the Bosco Quantum-Qubit competition; Technical Lead, Team Error404, National Robotics League 2025.

EXPERIENCE

FC Barcelona Academy World Cup, Spain – Team India *[Mon 'YY]*
- Selected to represent Team India, chosen from ~100 candidates nationally (1 of 2 from Kolkata); earned a starting XI place after a 5-day selection camp in Mumbai.
- Competed against national academy sides from Colombia, the Philippines, Belgium, Luxembourg, Spain and Ecuador.

Mohammedan Sporting Club Football Academy *[Mon 'YY] – Present*
- Training and playing with the [U19] squad of Mohammedan SC, an I-League club; nationally registered with the All India Football Federation (AIFF).
- Previously played for Southern Samity FC (U15) in the Reliance Foundation Youth Development League, and was selected for D'Souza Football Academy's AIFF National Youth League team.

Football Showcases: EXACT All-American and Connecticut Showcase *[Mon 'YY]*
- Selected for the EXACT All-American team to play in Spain, placing in the top 2–3% of athletes at the X1 Showcase Camp.
- Scored two goals at a Connecticut showcase against international competition, in front of coaches from Brown, UConn and Sarah Lawrence; received written praise from the head coach of Brown University.

Dalhousie Institute Football Club *[Mon 'YY] – Present*
- Youngest player fielded in adult city-wide tournaments, including the CC&FC Invitational 7s (U25); winner, Calcutta Parsee Club U17 Tournament.

Plaksha University – Young Technology Scholars Program *May '25 – Jun '25*
- Completed the Young Technology Scholars program at Plaksha University covering [topics, e.g. AI, robotics, design].

Sci-spire: Co-Founder and Core Team *2025 – 2027*
- Co-founded the school's first intra-school science competition, in partnership with the NGO Disappearing Dialogues, to bring robotics projects to the East Kolkata Wetlands.
- Working on Brain Rain, MHSI's annual robotics fest.

School Sports Captain & Football Captain, MHSI *2023 – 2027*
- Founded and captained the school football team, building it from a losing record to national-level TAISI competition and consistent wins; scored the school's first tournament goals.
- Roles held: School Sports Captain (2026–27); House Vice-Captain & Football Society Captain (2025–26); Football Society Vice-Captain (2023–25); Student Council Member (2023–24); Event Head, Showcase '26.

All Bengal Women's Union: Girls' Football Coach *[Mon 'YY] – [Mon 'YY]*
- Coached girls' football for ~180 hours over 7 months; helped organise and distribute free jerseys, shorts and boots so participants could play.

HONORS AND AWARDS

Winner, Jefferson Junior Debate *Apr '25*
U.S. Consulate and iLEAD

Special Mention, La Martiniere Calcutta MUN *2024*
UN General Assembly Fourth Committee (SPECPOL)

Silver Award, Cambridge Upper Secondary Science Competition *2023*

Top 6 of 350+ Teams, IIT-Kharagpur Young Innovators Programme *[Mon 'YY]*

Bronze Medal, Plaksha Young Creators League *[Mon 'YY]*
India-wide Top 50

Honorary Award, Goethe-Institut Pune Youth Leaders Congress *[Mon 'YY]*
Presented by a German Consulate representative

Winner, Calcutta Parsee Club U17 Football Tournament *[Mon 'YY]*

Individual Champion, MHSI Sports Day *[Mon 'YY]*
1st in Inter-House Football (15 goals in 3 games), Kho-Kho and Badminton; 2nd in Chess

Learner Attribute Award and Certificate of Achievement (Computer Science and French) *2023 – 24*
Modern High School International

Special Achievement Award for Sporting Pursuit *[Mon 'YY]*
[School Name]

INTERESTS AND EXTRACURRICULAR ACTIVITIES

Model UN: Head Delegate & Co-Chair *2022 – 2026*

- Head Delegate, SSAMUN (2025–26); Co-Chair of the UNEP Committee at JOAN MUN 2022 (St. Joan of Arc International School).
- Working to establish MHSI's first Model UN team and build a debate culture among juniors.

Trinity College London: Drum Kit, Grade 4 (Distinction) *[Mon 'YY]*

- Passed Grade 4 with Distinction after 6 years of training; performed in the Calcutta School of Music Students' Concerts (Nov 2023, May 2025).

School Band: Drummer *[Mon 'YY] – Present*

- Performed at the ShowCASe closing ceremony, Exuberance, Bosco Fest 2026 and La Concurrence 2024, and in the foundation course show 'The River Remembers'.

Guitar *[Mon 'YY] – Present*

- Self-taught guitarist for 1.5 years, steadily taking on harder pieces and playing in group settings to connect people through music.

MHSI Radio and Community Service *[Mon 'YY] – Present*

- Helped with the MHSI radio, speaking on air; helped raise funds for the Cuddles Foundation through bake sales; House Vice-Captain, gathering junior students' opinions to drive positive change.